

Multi-Person Skeleton–IMU Alignment Pipeline for Re-Identification

Course Project Final Report

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Abstract

We present a multi-person skeleton–IMU alignment pipeline that links wearable inertial streams to visible person tracks via motion-based cross-modal matching. To address data scarcity and accumulated error—the two core obstacles in practical deployment—we evaluate synthetic IMU generation, adaptive temporal sampling, and history-based decision making. Experiments show that no single synthetic backend dominates both acceleration and gyroscope fidelity, motivating a hybrid route; that adaptive slicing improves matching on large corpora but does not generalize to smaller datasets; and that history-based matching with greedy margin-based assignment substantially outperforms both per-frame and Hungarian baselines.

1. Introduction

Long-term multi-person monitoring requires more than detecting people in isolated frames. In indoor care, rehabilitation, sports, and human–computer interaction scenarios, the system must maintain identity over time so that motion and behavior statistics can be attributed to the correct individual. Modern multi-object tracking and pose-estimation systems have made substantial progress in this direction, from detection-association trackers such as ByteTrack [15] and FairMOT [14] to real-time whole-body pose systems such as AlphaPose [1] and video-based 3D motion recovery methods such as WHAM [8]. Nevertheless, camera-only identity is inherently fragile in the settings that motivate this project. Occlusion, close interaction, similar clothing, missed detections, and viewpoint changes can fragment a person’s visual trajectory or cause identity switches. These failure modes are especially damaging for long-term monitoring because a short visual ambiguity can corrupt the identity history accumulated over many windows.

Wearable inertial measurement units (IMUs) provide a complementary source of identity-bearing motion. An IMU stream is tied to a physical device and remains available

when the wearer is visually occluded, outside the field of view, or visually similar to another person. This property has made inertial sensing useful for human motion capture, including IMU-only pose recovery [2, 13] and visual-inertial human pose estimation [6, 10, 12, 16]. However, IMUs do not directly reveal where the wearer appears in an external camera. The camera observes people but not device IDs, while the IMU observes device-specific motion but not image-space location. The central problem is therefore not simply pose estimation from either modality; it is cross-modal identity assignment between visible skeleton motion and wearable sensor motion.

This report studies a skeleton–IMU alignment pipeline for multi-person re-identification. Given a video stream and synchronized IMU streams, the system converts video into tracked skeleton windows, converts real or synthetic IMU into synchronized sensor windows, and compares the two motion representations to infer which visible person corresponds to each device. We focus on two practical obstacles. Datasets that combine multi-person RGB video, reliable identity labels, and synchronized wearable IMU are much scarcer than pure video datasets or motion-capture corpora. At the same time, every upstream module—detection, tracking, pose estimation, sensor preprocessing, and temporal synchronization—injects noise that misaligns the two embedding manifolds, so matching accuracy is not determined by the matcher alone.

Synthetic data is a natural response to data scarcity, but it must be handled carefully. Parametric body models and motion repositories such as SMPL-X [7] and AMASS [4] make it possible to convert structured human motion into rendered visual data or sensor-like signals, and synthetic-human datasets such as SURREAL [11] demonstrate the value of rendering mocap-driven human data for vision tasks. For skeleton–IMU alignment, however, synthetic data must preserve both visual motion and inertial dynamics. A visually plausible rendered body is not sufficient if the corresponding acceleration or gyroscope stream has unrealistic timing, scale, or frequency structure. This motivates explicit validation of generated IMU signals before

using them as training or evaluation data.

This report makes three contributions toward practical skeleton-IMU alignment.

(1) **End-to-end pipeline.** We build a multi-person system that integrates detection, tracking, pose estimation, IMU preprocessing, and cross-modal matching, allowing us to study where and how error accumulates across stages.

(2) **Synthetic IMU generation.** We systematically evaluate generation methods and propose a hybrid backend that preserves both acceleration magnitude and gyroscope dynamics, reducing dependence on scarce paired recordings.

(3) **Accumulated-error mitigation.** We investigate two complementary strategies—adaptive temporal sampling and history-based decision making—and show that the latter, combined with greedy margin-based assignment, substantially improves matching stability over per-frame baselines.

2. Background

Skeleton-IMU alignment draws on three lines of prior work. Multi-object tracking and pose estimation convert RGB video into temporally associated human motion. Inertial and visual-inertial motion-capture methods show how sparse wearable sensors encode body dynamics and complement camera observations. Synthetic human-data generation, built on parametric body models and motion-capture corpora, offers a way to construct paired observations when real RGB-IMU recordings are limited.

2.1. Visual Tracking and Pose-Based Motion

Video provides the most direct observation of where people are in a scene. The standard computer-vision pipeline first detects people, associates detections over time, and optionally estimates 2D or 3D body pose for each track. Recent multi-object tracking methods improve association by using appearance embeddings, detection confidence, or joint detection-re-identification training. FairMOT formulates detection and re-identification as a joint problem [14], while ByteTrack shows that associating low-confidence detections can recover occluded or partially visible objects that would otherwise be discarded [15]. The HOTA metric further highlights why tracking quality cannot be reduced to detection alone: detection, association, and localization errors each affect the usefulness of a track [3].

For skeleton-IMU alignment, the video representation must describe motion rather than appearance alone. 2D pose systems such as AlphaPose provide temporally trackable keypoints for multiple people [1], and recent 3D human motion methods such as WHAM recover globally grounded motion from monocular video [8]. These advances make skeleton motion a meaningful visual-side signal for matching. However, they do not remove the identity problem. When people overlap, leave the field of view, or perform

similar actions, a tracker may split one person into multiple tracklets or swap identities between people. The downstream matcher therefore receives useful but imperfect skeleton windows.

2.2. Wearable IMU and Visual-Inertial Motion

IMUs measure acceleration, angular velocity, and orientation in the local sensor frame. They are lightweight, inexpensive, and robust to visual occlusion, which makes them attractive for long-term monitoring. IMU-only motion-capture systems such as DIP and TransPose show that sparse inertial measurements can recover full-body pose or translation under suitable assumptions [2, 13]. Visual-inertial human-motion methods further demonstrate that cameras and IMUs contain complementary information. TotalCapture fuses multi-view video and inertial measurements for 3D pose estimation [10], and later work combines sparse IMUs with a moving camera for in-the-wild pose recovery [12]. Recent methods such as DiffCap and Loose-Inertial-Poser continue this direction by using sparse IMUs, cameras, or loose-wear sensors for human motion capture [6, 16].

These systems show that inertial signals contain rich body-motion information, and that camera and IMU observations can be complementary. They also reveal a key distinction from the re-identification setting considered here. Most visual-inertial pose methods assume that the visual observation and the IMU stream are already associated with the same subject. In a multi-person identity-assignment problem, that association is exactly what must be recovered.

2.3. Synthetic Motion and Sensor Data

Paired RGB-IMU data is expensive to collect because it requires camera calibration or synchronization, wearable sensors, subject consent, and reliable identity labels. This is particularly limiting for multi-person settings, where the number of possible assignments grows with the number of people and devices. Synthetic data offers a way to increase coverage by starting from structured body motion and generating multiple observation modalities. Parametric body models such as SMPL-X provide a consistent surface and kinematic representation for human motion [7]. AMASS aggregates many motion-capture datasets into a unified body-model format, making it a useful source of motion sequences for synthesis [4]. In the vision literature, SURREAL shows that mocap-driven rendered humans can provide large-scale supervision for human analysis tasks [11].

For skeleton-IMU alignment, synthetic data is useful only if the generated modalities remain consistent with the same underlying body motion. Rendered video can expose a tracker or pose estimator to controlled visual observations, while synthetic IMU can expose the matcher to sensor-like

streams from the same motion source. However, inertial simulation introduces additional modeling choices that are not present in visual rendering alone. Finite differencing, coordinate transforms, smoothing, sensor placement, and simulated hardware noise can all change acceleration and gyroscope statistics. Tools such as WIMUSim explicitly model wearable-IMU variability [5], but the literature still suggests that generated IMU should be validated against real sensor measurements before it is used as a substitute for paired data.

3. Problem Setup and Pipeline Overview

We next define the skeleton-IMU assignment problem and summarize the pipeline assumed by the later method and experiments. The setup fixes notation, clarifies the system boundary, and explains why synthetic IMU validation and temporal window construction are evaluated separately from the final matcher.

3.1. Task Formulation

Let a synchronized recording contain a set of visual person tracks and a set of wearable IMU streams. For a temporal window, the video branch produces skeleton-motion candidates $\mathcal{V} = \{v_1, \dots, v_N\}$, and the sensor branch produces IMU windows $\mathcal{I} = \{i_1, \dots, i_M\}$. The goal is to infer which visual track and which wearable device describe the same person’s motion. Equivalently, the system estimates an assignment between visual tracks and IMU streams from cross-modal motion evidence.

In the learning-based route studied in this report, a skeleton encoder maps v_n to a motion embedding, an IMU encoder maps i_m to a sensor embedding, and the matcher scores candidate pairs by similarity. The output can be interpreted at two levels. Window-level decisions measure whether a single skeleton-IMU pair is ranked correctly, while grouped decisions measure whether the correct wearable is selected among multiple visible people. The grouped setting is closer to the intended application because multi-person identity assignment requires choosing among competing candidates rather than classifying one isolated pair.

3.2. Pipeline Overview

The full pipeline is organized as modality-specific processing followed by cross-modal matching. On the video side, the system detects and tracks people, estimates body pose, and converts each person track into a temporal skeleton sequence. On the IMU side, the system preprocesses real wearable streams or generates synthetic IMU from structured body motion. When paired real video-IMU data is unavailable, the data-construction stage can synthesize both rendered video and IMU streams from shared motion sources. Both modalities are then sliced into synchronized

windows and encoded into representations that can be compared by a matcher. Figure 1 shows the overall data flow and highlights the two components evaluated in this report.

The source design separates the high-level pipeline from the specific tracker, pose estimator, and matching backend. One route uses a tracker and 3D pose estimator with DeSPITE-style skeleton and IMU encoders. Another route uses a 2D tracking and pose stack, such as ByteTrack and AlphaPose, followed by skeleton-motion and IMU encoders. The system design also allows a physics-based matching route that compares motion cues derived from skeleton trajectories and IMU trajectories directly. The current report does not introduce new quantitative comparisons among these full-system variants. Instead, it evaluates two components that affect the quality of the matcher input: synthetic IMU generation and temporal window construction.

3.3. Challenges and Design Implications

3.3.1. Challenge 1: Data Scarcity

Paired video-IMU data is scarce and expensive to collect. Without sufficient training pairs, the matcher cannot learn robust cross-modal embeddings.

Design response. We reduce dependence on real paired data by generating synthetic IMU from structured body motion and validating it against real sensor measurements before downstream use.

3.3.2. Challenge 2: Accumulated Error

Matching accuracy is not determined by the matcher alone. Upstream noise—detection miss, tracking fragmentation, pose jitter, sensor bias, drift, and temporal misalignment—accumulates and misaligns the two embedding manifolds.

Design response. We mitigate accumulated error through two mechanisms: adaptive temporal sampling selects more informative windows to reduce ambiguous inputs, and history-based decision making reinforces high-confidence matches across frames while gating out noisy frames via a margin-based threshold.

4. Method

4.1. Synthetic Data Construction

4.1.1. Shared Generation Objective

The purpose of synthetic data construction is to reduce dependence on scarce paired video-IMU data. Given a structured body-motion representation, the synthetic data stage should produce synchronized visual and inertial observations that describe the same underlying motion. The rendered-video component supplies the video-side observation, while the synthetic-IMU component supplies the wearable-sensor observation. Keeping both modalities tied

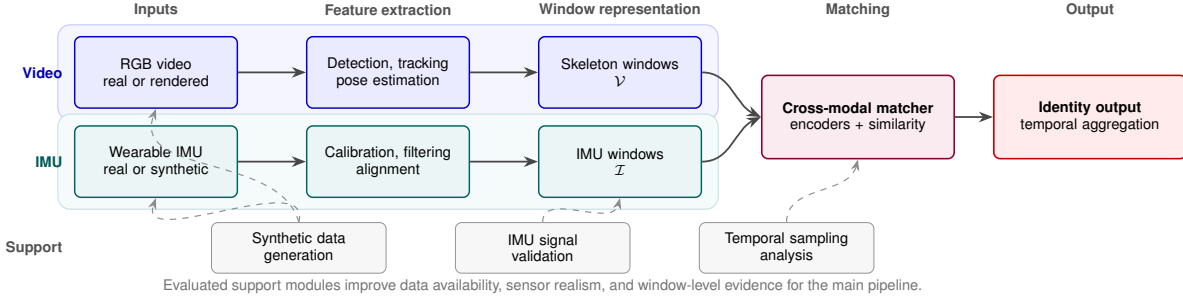


Figure 1. Overall skeleton-IMU alignment pipeline. The main path converts RGB video and wearable IMU streams into synchronized skeleton and inertial windows, scores cross-modal motion similarity, and aggregates window-level evidence into identity assignments. Synthetic data generation, IMU validation, and temporal sampling analysis support the main system.

to the same motion source is important for controlled skeleton-IMU alignment experiments.

4.1.2. Rendered Video Generation

The rendered-video component synthesizes photorealistic multi-person motion clips by driving textured human body models with motion-capture trajectories inside a controlled 3D scene. Its output is a set of synchronized RGB frames, instance segmentation masks, and 2D bounding-box annotations that can be fed directly into downstream pose estimators and trackers.

Motion source and body models. Body motion is sourced from the AMASS dataset <https://amass.is.tue.mpg.de/>, which aggregates SMPL-X parameter sequences from multiple motion-capture collections. Each AMASS .npz file provides per-frame body poses and root translations at the original capture frame rate. Before rendering, a sequence is resampled to the target video frame rate (30 fps) and its duration is standardized to 450 frames (15 s) by holding the last pose when the original motion is shorter.

Human avatars are built from the SynBody asset library, which provides clothed SMPL-XL models exported as rigged .fbx files. For each sequence, the pipeline samples distinct actor identities and assigns each a random motion, a random ground-plane location, and a random facing direction. Long sequences use 7 actors per scene; shorter sequences use 4–6 actors. A conservative placement policy (max radius 2.5 m, minimum inter-actor distance 0.8 m) prevents inter-penetration while still allowing natural occlusions.

Scene and lighting. The scene is composed inside headless Blender 3.6.4 via the XRFeitoria 0.6.3 Python API. A ground plane receives a procedurally tiled diffuse texture, and environment lighting is supplied by a randomly selected HDRI dome. No additional mesh props or dynamic objects

are introduced, so the visual complexity is dominated by the actors themselves and their mutual occlusion patterns.

Camera configuration. Eight static camera presets surround the scene at a fixed height of approximately 3.0 m: front, back, left, right, and the four diagonal corners. All cameras look at the scene center ($z \approx 0.8$ m), producing a mix of eye-level and mild overhead viewpoints. Only one camera is rendered per sequence in the current configuration, but the preset system is designed to allow multi-view extensions without changing the asset layout.

Rendering engine and output. Rendering is performed with Blender’s Cycles engine using OptiX acceleration on NVIDIA RTX 4090 GPUs. The default settings are 1280×720 resolution, 128 samples per pixel, and an unbounded per-frame time limit. Each GPU processes one sequence independently; eight GPUs run in parallel so that a 10-sequence batch finishes in roughly the same wall-clock time as a single sequence.

For every frame the engine produces:

- an RGB image (PNG) in the `rgb/` folder;
- an instance mask pass in which each pixel stores the actor’s persistent stencil ID, used for exact 2D bounding-box extraction without threshold heuristics;
- per-actor camera-space bounding boxes saved as JSON for direct dataloader ingestion.

After the render completes, the pipeline automatically stitches the PNG frames into raw and bounding-box-overlay MP4 videos for quick visual validation.

Synchronization with IMU. Rendered video and synthetic IMU share the same underlying motion source. The AMASS motion that drives the body model is also fed into the IMU synthesis backend (Sec. 4.1.3) before rendering starts. Both modalities therefore cover identical frame ranges at identical world-space timings; any mismatch is limited to resampling jitter and the IMU sensor’s

local-frame coordinate transform, not to independent motion sources.

Quality-control checks. Several filters and post-hoc checks keep the rendered output within usable bounds.

- **Motion pre-filtering:** Motions with excessive vertical variation (z -range > 0.3 m) or extreme spine angles are rejected before assignment to an actor, because they tend to produce anatomically implausible poses when retargeted to the SMPL-XL rig.
- **Ground penetration correction:** After motion resampling, the minimum pelvis height across all frames is computed. If it is below a small safety margin (0.05 m), a global vertical offset is added to every frame so that no actor clips through the ground plane.
- **Visual inspection:** For each rendered sequence, side-by-side raw and bounding-box-overlay videos are generated automatically. A quick scan of these videos reveals missing actors, floating limbs, or camera misalignment without requiring frame-by-frame mask analysis.
- **Compatibility check:** The rendered clips are validated against the downstream tracker and pose-estimator input requirements (fixed resolution, correct color channels, consistent actor IDs across frames).

These checks do not guarantee perfect realism, but they eliminate the most common synthetic-artifact classes that would otherwise break downstream skeleton extraction or IMU alignment.

4.1.3. Synthetic IMU Generation

The synthetic IMU component converts structured body motion into wearable-sensor streams with the same column layout as the processed real IMU data. Its input is a body-motion sequence, such as processed SMPL-X parameters or AMASS motion, together with a target sensor location. Its output is a synchronized CSV stream containing orientation, acceleration, gyroscope, and optional magnetometer fields. This shared output contract allows downstream slicing and matching code to consume real and synthetic IMU through the same interface.

SMPL-X to virtual sensor motion. For each motion sequence, the body model is evaluated frame by frame to recover joint or mesh trajectories. A target body location, such as a lower arm sensor, is then mapped to a virtual sensor trajectory. Depending on the backend, the sensor trajectory can be represented by a joint position, a nearby mesh vertex, or a full forward-kinematic body model. The synthesis backend converts this motion into local-frame inertial signals.

Candidate synthesis backends. The implementation keeps several synthesis routes behind the same output in-

terface. `naive_kinematics` differentiates joint trajectories and rotates gravity into the sensor frame. The vertex-based routes, inspired by DiffCap [6] and Loose-Inertial-Poser [16], run SMPL-X forward kinematics, track a mesh vertex near the target sensor, and compute acceleration by finite differences with different smoothing levels. `ssl_imu_style` applies spline smoothing before differentiation. `wimusim` converts SMPL-X parameters into a WIMUSim [5] B-D-P-H model and simulates gyroscope and orientation through forward kinematics. Noise-augmentation variants add Gaussian noise, random-walk drift, or bias drift to a clean synthetic signal.

Hybrid candidate. Because acceleration and gyroscope quality can favor different backends, the pipeline also defines a hybrid candidate. The hybrid route fuses acceleration columns from one method with gyroscope and quaternion columns from another method after temporal alignment. This is implemented by a reusable merger that accepts per-method CSV files and writes a single synchronized IMU stream. The empirical comparison used to choose the default synthetic stream is reported in Sec. 6.2.

HuMoGen integration. For large-scale synthetic data generation, hybrid IMU can be pre-computed directly from AMASS `.npz` motion files before rendering. The pre-computation pipeline loads SMPL-X parameters, computes candidate acceleration and gyroscope streams, merges them, and resamples the result to the target IMU frequency. During HuMoGen rendering, the rendering script checks for the pre-computed CSV and aligns its length with the rendered clip by truncation or padding. If the pre-computed file is absent, the pipeline falls back to the original Blender-trajectory synthesis path.

4.2. Matching Pipeline

4.2.1. Window Definition

The matcher operates on synchronized temporal windows rather than full sequences. Each window contains a skeleton segment from one tracked person and an IMU segment from one candidate wearable stream. All downstream experiments in this report keep the model input length fixed at $L = 24$ frames. The fixed input length isolates the effect of sampling policy from the effect of model architecture.

4.2.2. Embedding Window and Candidate Group

An embedding window is the frame interval used to form one skeleton or IMU embedding. A candidate group contains the people or tracks considered together during matching. Grouped evaluation is important because the target application is multi-person identification: the system must distinguish the correct wearer from other people present in the same scene, not simply classify one isolated window. This

is why grouped mean accuracy receives more weight than Top-1 in the composite selection metric.

4.2.3. Chunk-Level Aggregation

The project source material also distinguishes an embedding window from a chunk. A chunk is a set of consecutive embeddings used for one video-level matching decision. Longer chunks can provide more temporal evidence, but they can also accumulate pose and tracking errors. The current report does not introduce new chunk-size results; instead, it uses this observation to motivate why temporal-window construction deserves separate analysis.

4.3. Accumulated Error Mitigation

4.3.1. Adaptive Temporal Sampling

Fixed-Window Baseline The baseline uses a fixed temporal grid. For each sequence, windows start at

$$\mathcal{S}_{\text{fixed}} = \{0, s, 2s, \dots\}, \quad (1)$$

where the window length is $L = 24$ and the stride is $s = 16$. This strategy is simple and reproducible, but it samples calm and high-motion segments at the same density.

Motion-Energy Adaptive Slicing The first adaptive strategy uses motion energy to vary the sampling stride. For each frame, the slicer computes temporal change from IMU features and skeleton trajectories, normalizes these signals, and averages them into a frame-level score. Candidate window scores are then obtained by averaging frame scores over each length- L window.

For a candidate window score e , the next stride is selected by quantile thresholds:

$$\Delta t = \begin{cases} \text{min_stride}, & e \geq q_1, \\ \text{mid_stride}, & q_0 \leq e < q_1, \\ \text{max_stride}, & e < q_0. \end{cases} \quad (2)$$

Thus, high-motion regions are sampled more densely and low-motion regions more sparsely. In the experiments, we evaluate two hyperparameter groups: g1 uses min_stride=8, max_stride=24, and $(q_0, q_1) = (0.40, 0.75)$; g2 uses min_stride=6, max_stride=24, and $(q_0, q_1) = (0.35, 0.70)$.

Fixed-Plus-Energy Slicing The second adaptive strategy is designed to be more conservative. It preserves the fixed stride-16 grid as coverage, then adds extra windows centered on high-motion candidate starts. This keeps the stability of fixed sampling while increasing representation of informative motion segments. We evaluate two settings: g1 adds 30% extra windows with a minimum gap of 8 frames; g2 adds 50% extra windows with a minimum gap of 4 frames.

Selection Metric To rank temporal sampling methods, we use a composite score:

$$\text{Composite} = 0.7 \cdot \text{GroupedMeanAcc} + 0.3 \cdot \text{Top1}. \quad (3)$$

Grouped mean accuracy is weighted more heavily because the target application requires robust matching among multiple candidate people, while Top-1 accuracy captures single-window classification quality.

4.3.2. History-Based Decision Making

Independent per-frame matching is fragile: a single noisy embedding can flip the assignment for that frame. We therefore accumulate evidence across time using a temporally decayed score matrix.

Window Embedding and Similarity Let $\mathbf{x}_i^{(t)} \in \mathbb{R}^d$ denote the IMU embedding for sensor i and $\mathbf{y}_j^{(t)} \in \mathbb{R}^d$ the skeleton embedding for track j , both extracted from a temporal window centered at frame t . The cosine similarity between them is

$$s_{ij}^{(t)} = \frac{\mathbf{x}_i^{(t)} \cdot \mathbf{y}_j^{(t)}}{\|\mathbf{x}_i^{(t)}\| \|\mathbf{y}_j^{(t)}\|}, \quad (4)$$

which is collected into a similarity matrix $\mathbf{S}^{(t)} \in \mathbb{R}^{N_{\text{imu}} \times N_{\text{pred}}}$.

SignedVote Accumulation Rather than assigning from $\mathbf{S}^{(t)}$ alone, we maintain an accumulated score matrix $\mathbf{H}^{(t)}$ that aggregates evidence over time with exponential decay:

$$\mathbf{H}^{(t)} = (1 - \lambda) \mathbf{H}^{(t-1)} + \mathbf{S}^{(t)}, \quad (5)$$

where $\lambda \in [0, 1]$ is a forgetting rate. $\lambda = 0$ yields pure cumulative summation; larger values discard older evidence faster. The frame- t assignment is derived from $\mathbf{H}^{(t)}$, not from $\mathbf{S}^{(t)}$.

Greedy Assignment with Margin Ordering Greedy assignment operates on $\mathbf{H}^{(t)}$ as follows. For each IMU i , let

$$j_i^* = \arg \max_j H_{ij}^{(t)}, \quad v_i^{(1)} = H_{ij_i^*}^{(t)}, \quad (6)$$

be the best matching track and its score. Let

$$v_i^{(2)} = \max_{j \neq j_i^*} H_{ij}^{(t)} \quad (7)$$

be the second-best score. The *margin* for IMU i is

$$m_i = v_i^{(1)} - v_i^{(2)}. \quad (8)$$

The algorithm sorts all IMUs by descending margin, then iteratively assigns each IMU to its best available track.

Pseudocode is given in Alg. 4.3.2. Margin ordering ensures that the most confident matches are committed first; ambiguous IMUs (low m_i) are decided only after their competing tracks have been removed from the candidate pool.

[t] [1] Accumulated score matrix $\mathbf{H} \in \mathbb{R}^{N_{\text{imu}} \times N_{\text{pred}}}$ Assignment $\mathcal{A} : \{1, \dots, N_{\text{imu}}\} \rightarrow \{-1, 1, \dots, N_{\text{pred}}\}$ $\mathcal{A} \leftarrow \emptyset$, $\mathcal{T} \leftarrow \{1, \dots, N_{\text{pred}}\}$ each IMU i $j_i^* \leftarrow \arg \max_j H_{ij}$ $v_i^{(1)} \leftarrow H_{ij_i^*}$, $v_i^{(2)} \leftarrow \max_{j \neq j_i^*} H_{ij}$ $m_i \leftarrow v_i^{(1)} - v_i^{(2)}$ Sort IMUs by m_i in descending order IMU i in sorted order $\mathcal{T} \neq \emptyset$ $j^* \leftarrow \arg \max_{j \in \mathcal{T}} H_{ij}$ $\mathcal{A}[i] \leftarrow j^*$, $\mathcal{T} \leftarrow \mathcal{T} \setminus \{j^*\}$ $\mathcal{A}[i] \leftarrow -1$ unsigned $\mathcal{A}$

Hungarian Assignment The Hungarian algorithm finds the global optimum

$$\mathcal{A}^* = \arg \max_{\mathcal{A}} \sum_{i=1}^{N_{\text{imu}}} H_{i, \mathcal{A}(i)}, \quad (9)$$

where $\mathcal{A}$ ranges over all one-to-one assignments. When $N_{\text{imu}} = N_{\text{pred}} = 2$, the solution space has size 2 and Greedy and Hungarian are equivalent. For $N \geq 3$, the two can diverge: Hungarian maximizes the *total* score, whereas Greedy maximizes the *minimum margin* implicitly through its ordering.

Hard Threshold Strategy Equation (5) updates history unconditionally, which means a single noisy frame can corrupt the accumulated scores. The Hard Threshold strategy gates the update on a per-IMU confidence test.

Using the margin defined in Eq. (8), the update rule becomes

$$\mathbf{H}^{(t)} = \begin{cases} \mathbf{H}^{(t-1)}, & \text{if } m_i < \tau \text{ for any } i \in \{1, \dots, 2\text{-person}\} \\ (1 - \lambda) \mathbf{H}^{(t-1)} + \mathbf{S}^{(t)}, & \text{otherwise.} \end{cases} \quad (10)$$

The first branch is the *preserve* action: the entire score matrix is frozen when *any* IMU shows low confidence. The threshold $\tau = 0.7$ is selected by grid search. Two alternative low-confidence actions are also explored:

- **skip**: apply decay but do not add the new similarity, i.e. $\mathbf{H}^{(t)} = (1 - \lambda) \mathbf{H}^{(t-1)}$;
- **light_decay**: use a smaller decay for the affected row only.

The preserve action achieves the best empirical performance because it completely blocks noise injection.

Learned Confidence Estimator The hand-crafted margin m_i may not capture all cues of reliability. We therefore train a lightweight MLP $g_{\theta} : \mathbb{R}^6 \rightarrow [0, 1]$ that predicts the probability that the current frame is reliable enough to update history.

The input feature vector $\mathbf{f} \in \mathbb{R}^6$ for frame t is computed from $\mathbf{S}^{(t)}$ as follows:

$$f_1 = \frac{1}{N} \sum_{i=1}^N (\max_j S_{ij} - \max_{j \neq j_i^*} S_{ij}), \quad (\text{diagonal margin}) \quad (11)$$

$$f_2 = \max_i S_{ii}, \quad (\text{max diagonal}) \quad (12)$$

$$f_3 = \frac{1}{N} \sum_{i=1}^N (\sigma_i(i) - \max_{k \neq i} \sigma_i(k)), \quad (\text{softmax margin}) \quad (13)$$

$$f_4 = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N \sigma_i(j) \log \sigma_i(j), \quad (\text{entropy}) \quad (14)$$

$$f_5 = \frac{1}{N} \sum_{i=1}^N \max_j S_{ij}, \quad (\text{best score}) \quad (15)$$

$$f_6 = \frac{1}{N} \sum_{i=1}^N \max_{j \neq j_i^*} S_{ij}, \quad (\text{second-best score}) \quad (16)$$

where $\sigma_i(j) = \exp(S_{ij}) / \sum_k \exp(S_{ik})$ is the softmax distribution over tracks for IMU i , and $j_i^* = \arg \max_j S_{ij}$.

The network architecture is

$$g_{\theta}(\mathbf{f}) = \sigma(\mathbf{W}_3 \phi(\mathbf{W}_2 \phi(\mathbf{W}_1 \mathbf{f} + \mathbf{b}_1) + \mathbf{b}_2) + b_3), \quad (17)$$

where $\phi(\cdot) = \text{ReLU}(\cdot)$, $\mathbf{W}_1 \in \mathbb{R}^{32 \times 6}$, $\mathbf{W}_2 \in \mathbb{R}^{16 \times 32}$, $\mathbf{W}_3 \in \mathbb{R}^{1 \times 16}$, and dropout ($p = 0.2$) is applied after the first hidden layer. The model is trained with binary cross-entropy on ground-truth correctness labels derived from the 2-person Local branch (Fold 1 and Fold 2 for training, Fold 3 and Fold 4 for validation), achieving validation AUC = 1.000 and accuracy = 99.5%.

At inference, the update rule is

$$\mathbf{H}^{(t)} = \begin{cases} (1 - \lambda) \mathbf{H}^{(t-1)} + \mathbf{S}^{(t)}, & \text{if } g_{\theta}(\mathbf{f}^{(t)}) \geq \tau_{\text{ce}}, \\ (1 - \lambda) \mathbf{H}^{(t-1)}, & \text{otherwise,} \end{cases} \quad (18)$$

with $\tau_{\text{ce}} = 0.5$. The second branch (*skip*) decays history without adding new evidence. An alternative *preserve* branch ($\mathbf{H}^{(t)} = \mathbf{H}^{(t-1)}$) is discussed in Sec. 6.3.2.

5. Datasets

TotalCapture benchmark. TotalCapture [10] is used for both synthetic IMU validation and video-based matching experiments because it provides synchronized body motion, video, and wearable IMU measurements. A custom legacy two-person dataset is used as a smaller downstream matching test closer to the multi-person target setting. The report

does not use the custom dataset to draw conclusions about synthetic IMU fidelity because raw ground-truth IMU comparisons are only reported for the TotalCapture setup.

In-house collection. To address the scarcity of paired multi-person video-IMU data, we conducted two in-house capture sessions with wearable IMUs and overhead cameras. The first session served as a feasibility pilot: four 1.5-minute recordings covering random walking with overlaps, static poses, small-scale movements, and large-scale movements at a fixed point. The second session was a systematic collection of approximately 92 minutes of data involving 1, 2, 4, and 7 participants. Scenes included unobstructed and obstructed environments, and activities ranged from static poses and in-place motion to free walking and a long-duration interactive session in which participants rearranged chairs to clear the space, followed by guided daily actions (drinking, waving, closing doors, and other tasks displayed on a screen), as well as more vigorous sports such as shuttlecock kicking and basketball that introduced rapid limb motion and heavier occlusion. To mitigate sensor-specific bias, odd and even participant groups swapped IMUs between rounds, and recordings were split into training and held-out test sets. This custom dataset is used only for downstream matching experiments; it is not used to validate synthetic IMU fidelity because it lacks the paired mocap reference required for per-axis ground-truth comparison.

Synthetic TotalCapture IMU. To augment the scarce paired video-IMU data, we treat the SMPL-X parameters fitted to TotalCapture as ground-truth motion and re-synthesize IMU streams from them. The synthesis pipeline covers 37 sequences from subjects S1-S5 and two arm-mounted sensors (`arm.right` and `arm.left`), yielding 74 sensor-sequence pairs. We compared ten generation backends, including kinematic finite-difference methods, vertex-based second-derivative methods, a physics-forward pipeline, and hybrid combinations. Signal-level validation against real TotalCapture IMU uses per-axis Pearson correlation, magnitude RMSE, log-normalized PSD distance, peak-timing error, and one-second window-statistic RMSE. The hybrid candidate, which takes acceleration from `naive_kinematics` and gyroscope from `wimusim`, emerged as the strongest compromise across modalities and is used for downstream matching experiments.

Rendered video and paired IMU. Beyond re-synthesizing IMU from existing mocap, we also generate fully synthetic paired data by rendering SMPL-X animations as RGB video while simultaneously producing the corresponding IMU stream with the validated backend. The rendered clips provide synchronized RGB, mask,

and bounding-box outputs. However, manual inspection revealed three artifact classes that currently prevent their use in training: ground penetration during deep knee bends, floating or drifting root trajectories retargeted from treadmill-captured motions, and clothing-mesh self-penetration under large limb motions. Because of these issues the rendered video is withheld from the training set and used only for qualitative inspection and pipeline debugging. Once ground-contact enforcement, motion-quality filtering, and garment-mesh robustness are addressed, the rendered video can be combined with the validated synthetic IMU to form a large-scale fully paired dataset for multi-person Re-ID training.

6. Experiments

6.1. Evaluation Setup

The experiments evaluate two parts of the pipeline. First, synthetic IMU generation is evaluated at the signal level by comparing generated streams against raw TotalCapture IMU measurements. Second, temporal slicing is evaluated through downstream skeleton-IMU matching accuracy. This separation is important because a synthetic IMU backend can preserve sensor dynamics without necessarily improving the final matcher, and a slicing strategy can affect matching even when the underlying sensor stream is unchanged.

Matching setup. All temporal-slicing variants are trained for 15 epochs with seed 42. The compared slicing methods are `fixed`, `motion_energy_g1`, `motion_energy_g2`, `fixed_plus_energy_g1`, and `fixed_plus_energy_g2`. Top-1 accuracy is reported for individual window-level candidate decisions. Grouped mean accuracy is computed after forming candidate groups and measures whether the correct IMU is selected within the group. Because these two metrics are computed over different decision units, their absolute values should not be interpreted as directly comparable. The composite score is defined in Sec. 4 and weights grouped mean accuracy more heavily than Top-1.

6.2. Synthetic IMU Evaluation

Synthetic IMU is evaluated in two complementary ways. The broad comparison in Sec. 6.2.1 compares the newer candidate backends across multiple TotalCapture sequences and is used for method selection. The single-sample sanity check in Sec. 6.2.2 compares legacy or maintained generation paths on one TotalCapture sample and is used only to inspect signal behavior for the existing workflow.

6.2.1. Broad Method Comparison

The broad comparison evaluates generated IMU against real IMU on 37 TotalCapture sequences from subjects S1-

S5 and two sensors with ground truth, `arm_right` and `arm_left`. This yields 74 sensor-sequence pairs. Metrics include Pearson correlation of acceleration and gyroscope magnitude, magnitude RMSE, log-normalized PSD distance using Welch’s method, peak-timing error, and one-second window-statistic RMSE.

The broad comparison shows that no single backend dominates both acceleration and gyroscope fidelity. `naive_kinematics` gives the highest acceleration correlation, while `wimusim` gives the strongest gyroscope correlation and lowest gyroscope error. The vertex-based methods show that smoothing level matters for acceleration quality, and the noise variants stay close to their clean baseline rather than improving fidelity. Based on these results, the hybrid candidate combines acceleration from `naive_kinematics` with gyroscope and quaternion columns from `wimusim`.

6.2.2. Legacy Sample Sanity Check

The broad comparison above is separate from a single-sample sanity check for legacy or maintained generation paths. For this check, we use the TotalCapture `S1_freestyle3` sample and compare the generated `R.LowArm` stream against the raw `R.LowArm` IMU. The comparison covers magnitude correlation and RMSE for acceleration and gyroscope.

Table 3 compares three generation backends evaluated on the same sample. `naive_kinematics` best matches acceleration magnitude on this sample, with the highest acceleration correlation and lowest acceleration RMSE. `GlobalPose_origin`, the maintained legacy workflow, gives the best gyroscope correlation and substantially lower gyroscope RMSE than the other two backends. `HuMoGen_origin` is weaker on this sample. This single-sample check is useful for inspecting the existing workflow, but it is not used as the sole basis for broad method selection.

6.3. Accumulated Error Mitigation

6.3.1. Adaptive Temporal Sampling Results

TotalCapture Video Results Table 4 shows that `fixed_plus_energy_g1` achieves the best composite score on TotalCapture. Compared with the fixed baseline, it improves grouped mean accuracy from 0.9166 to 0.9648 and Top-1 accuracy from 0.0702 to 0.0748. This supports the view that preserving fixed-grid coverage while adding high-motion windows can improve matching in the TotalCapture workflow.

Custom Dataset Results On the custom dataset, the fixed baseline remains strongest by composite score (Table 5). The best adaptive Top-1 score is achieved by `motion_energy_g1`, but its grouped mean accuracy

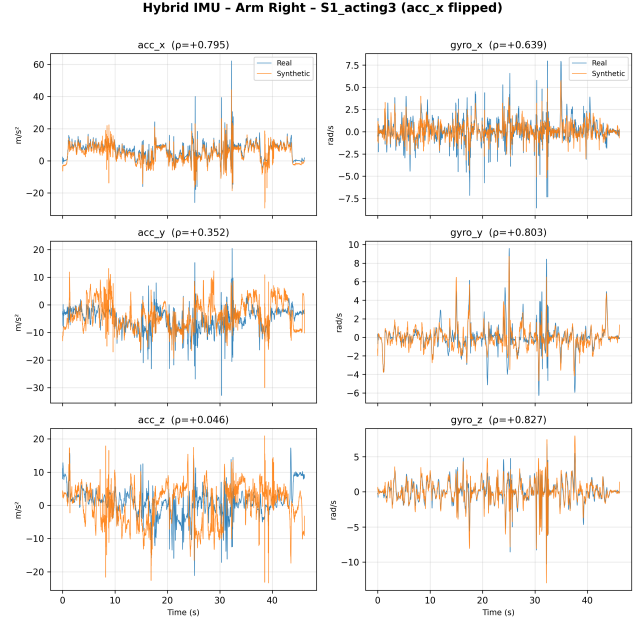


Figure 2. Hybrid IMU synthesis for `arm_right` on TotalCapture `S1_acting3`. Acceleration (left column) is taken from `naive_kinematics` and gyroscope (right column) from `wimusim`, demonstrating per-axis correlation after a simple sign correction on `acc_x`. The hybrid design is motivated by the observation that no single backend dominates both modalities: `naive_kinematics` captures acceleration well via wrist-position second derivatives, while `wimusim` yields more stable gyroscope estimates through joint-level forward kinematics.

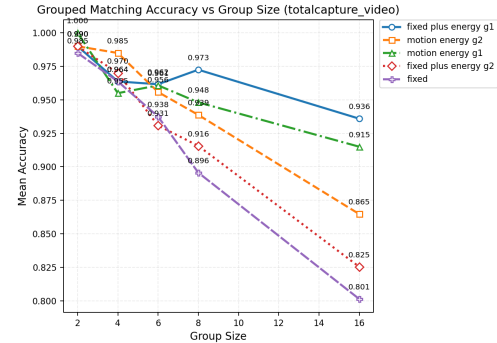


Figure 3. TotalCapture grouped matching accuracy across group sizes.

drops enough that its composite score remains below fixed. This suggests that adaptive sampling can improve isolated window-level discrimination while still hurting group-level matching robustness.

6.3.2. History-Based Matching Results

SignedVote Baseline Table 6 presents the per-frame accuracy on the 2-person merged data across six decay values.

Table 1. Overview of the synthetic IMU generation and augmentation methods in the broad comparison.

Method	Category	Description	Source
naive_kinematics	Baseline	Joint-angle finite-difference + gravity rotation.	In-house
vertex_diffcap	Vertex-based	SMPL-X vertex 2nd-order finite difference ($\text{smooth}_n = 2$).	DiffCap [6]
vertex_loose	Vertex-based	SMPL-X vertex finite difference ($\text{smooth}_n = 4$, stronger smoothing).	Loose-Inertial-Poser [16]
ssl_imu_style	Spline-based	Spline-interpolated 2nd derivative of vertex trajectory.	SSL-IMU [9]
wimusim	Physics FK	Full B-D-P-H forward kinematics with PyBullet physics.	WIMUSim [5]
noise_gaussian	Augmentation	Additive white Gaussian noise on a clean synthetic signal.	In-house
noise_random_walk	Augmentation	Cumulative random-walk drift.	In-house
noise_bias_drift	Augmentation	Gauss–Markov bias-drift process.	In-house

Table 2. Quantitative comparison of synthetic IMU methods on TotalCapture (37 sequences, 2 sensors). Bold = best per metric.

Method	Corr. $\uparrow$		RMSE $\downarrow$		PSD $\downarrow$		Peak (s) $\downarrow$		Win. RMSE $\downarrow$	
	Acc	Gyro	Acc	Gyro	Acc	Gyro	Acc	Gyro	Acc	Gyro
naive_kinematics	0.79$\pm$0.20	0.33 $\pm$ 0.12	3.68$\pm$6.82	7.70 $\pm$ 4.21	0.81 $\pm$ 0.38	1.65 $\pm$ 0.42	1.06 $\pm$ 3.61	1.35 $\pm$ 1.48	173.8 $\pm$ 1025.0	90.5 $\pm$ 82.2
vertex_diffcap	0.30 $\pm$ 0.15	0.32 $\pm$ 0.12	12.29 $\pm$ 7.21	7.74 $\pm$ 4.21	1.43 $\pm$ 0.41	1.65 $\pm$ 0.42	0.70 $\pm$ 0.83	1.35 $\pm$ 1.48	296.0 $\pm$ 974.3	90.5 $\pm$ 82.2
vertex_loose	0.54 $\pm$ 0.19	0.32 $\pm$ 0.12	4.92 $\pm$ 2.79	7.74 $\pm$ 4.21	0.58$\pm$0.17	1.65 $\pm$ 0.42	0.26$\pm$0.54	1.35 $\pm$ 1.48	35.5$\pm$36.0	90.5 $\pm$ 82.2
ssl_imu_style	0.21 $\pm$ 0.13	0.32 $\pm$ 0.12	21.95 $\pm$ 15.02	7.74 $\pm$ 4.21	1.36 $\pm$ 0.41	1.65 $\pm$ 0.42	0.81 $\pm$ 0.92	1.35 $\pm$ 1.48	1186.6 $\pm$ 4787.8	90.5 $\pm$ 82.2
wimusim	0.25 $\pm$ 0.16	0.86$\pm$0.16	8.67 $\pm$ 6.40	0.89$\pm$0.69	0.96 $\pm$ 0.37	0.66$\pm$0.39	0.38 $\pm$ 0.82	0.17$\pm$0.40	152.4 $\pm$ 349.1	1.87$\pm$2.86
noise_gaussian	0.30 $\pm$ 0.15	0.32 $\pm$ 0.12	12.29 $\pm$ 7.21	7.74 $\pm$ 4.21	1.43 $\pm$ 0.41	1.65 $\pm$ 0.42	0.70 $\pm$ 0.83	1.35 $\pm$ 1.48	296.0 $\pm$ 974.4	90.5 $\pm$ 82.2
noise_random_walk	0.30 $\pm$ 0.15	0.31 $\pm$ 0.12	12.37 $\pm$ 7.18	7.78 $\pm$ 4.20	1.43 $\pm$ 0.41	1.65 $\pm$ 0.42	0.67 $\pm$ 0.81	1.36 $\pm$ 1.51	297.7 $\pm$ 974.7	90.8 $\pm$ 82.4
noise_bias_drift	0.30 $\pm$ 0.15	0.31 $\pm$ 0.12	12.33 $\pm$ 7.20	7.77 $\pm$ 4.20	1.43 $\pm$ 0.41	1.65 $\pm$ 0.42	0.71 $\pm$ 0.87	1.35 $\pm$ 1.52	296.8 $\pm$ 975.1	90.7 $\pm$ 82.3

Table 3. Synthetic IMU sanity check on TotalCapture Sl_freestyle3, sensor R_LowArm. Higher correlation is better; lower RMSE is better.

Method	Acc Corr.	Gyro Corr.	Acc RMSE	Gyro RMSE
GlobalPose.origin	0.6250	0.3524	4.4776	5.5456
naive_kinematics	0.9751	0.2692	1.1852	8.2852
HuMoGen.origin	0.3004	0.2689	5.5185	8.2876

Table 4. TotalCapture video results. Adaptive fixed-plus-energy improves both grouped accuracy and Top-1 over the fixed baseline.

Method	Windows	Top-1	Grouped	Composite
fixed_plus_energy_g1	14571	0.0748	0.9648	0.6978
motion_energy_g2	14900	0.0953	0.9469	0.6914
motion_energy_g1	12058	0.0705	0.9558	0.6902
fixed_plus_energy_g2	16807	0.0808	0.9264	0.6727
fixed	11165	0.0702	0.9166	0.6627

Table 5. Custom dataset results. Fixed-window slicing remains best by composite score.

Method	Windows	Top-1	Grouped	Composite
fixed	2400	0.0152	0.3553	0.2533
motion_energy_g2	3180	0.0158	0.3100	0.2217
fixed_plus_energy_g2	3624	0.0163	0.3010	0.2156
motion_energy_g1	2524	0.0252	0.2943	0.2135
fixed_plus_energy_g1	3144	0.0125	0.2471	0.1767

Pure accumulation ($\lambda = 0$) achieves the highest mean accuracy of 70.0%, but the optimal decay varies strongly by session: Fold 1 reaches perfect accuracy (100.0%), whereas Fold 2 drops to 30.8% because erroneous early assignments

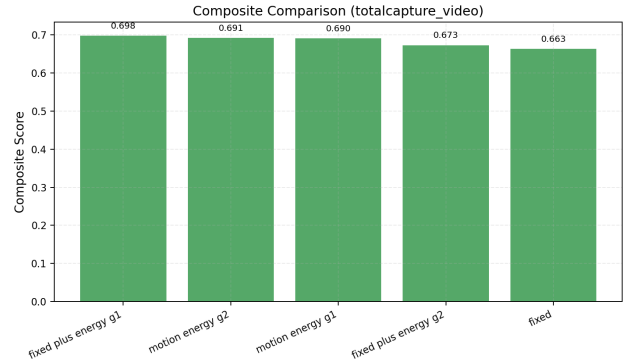


Figure 4. TotalCapture composite score comparison.

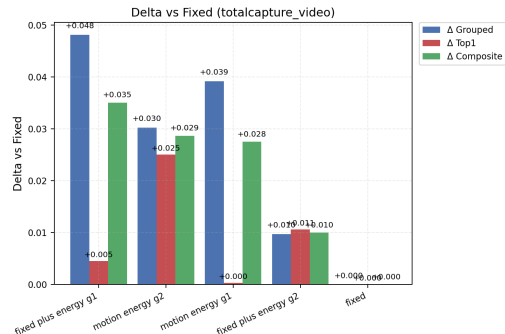


Figure 5. TotalCapture metric deltas relative to fixed-window slicing.

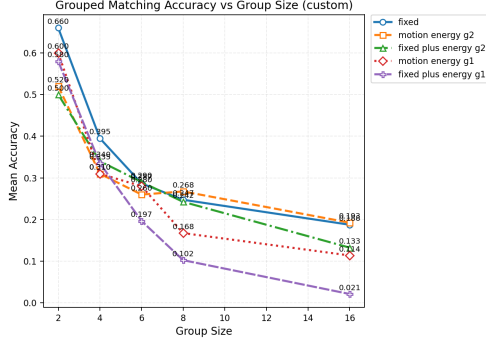


Figure 6. Custom dataset grouped matching accuracy across group sizes.

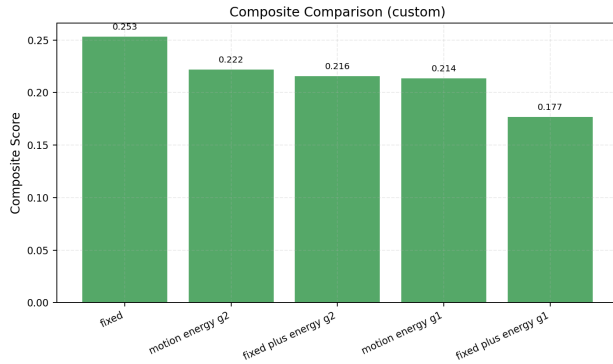


Figure 7. Custom dataset composite score comparison.

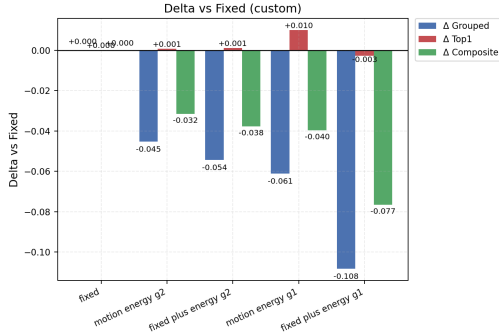


Figure 8. Custom dataset metric deltas relative to fixed-window slicing.

are locked in. A moderate decay ($\lambda \geq 0.20$) stabilizes Fold 2 at $\sim 49\%$ while only modestly hurting Fold 1.

On the unmerged data (Table 7), where skeleton track identities are fragmented, the optimal decay shifts to $\lambda = 0.02$ (54.1%). Pure accumulation is particularly harmful for Fold 2 (22.0%), again confirming that noisy initial assignments must be forgotten quickly.

Table 6. SignedVote results on the 2-person merged data (syn test, greedy, %).

Fold	$\lambda=0.00$	0.02	0.05	0.10	0.20	0.50
1	100.0	75.5	68.8	68.3	66.6	65.9
2	30.8	33.1	43.8	47.6	49.0	49.7
3	81.4	52.5	44.9	44.4	43.8	44.5
4	67.6	57.1	56.5	60.1	62.5	63.1
Mean	70.0	54.6	53.5	55.1	55.5	55.8

Table 7. SignedVote results on the 2-person unmerged data (syn test, greedy, %).

Fold	$\lambda=0.00$	0.02	0.05	0.10	0.20	0.50
1	44.5	45.9	45.9	43.9	44.9	44.3
2	22.0	64.3	64.1	59.9	55.5	54.1
3	53.4	54.5	51.2	50.3	51.0	50.3
4	54.3	51.5	51.9	53.0	51.9	50.7
Mean	43.6	54.1	53.3	51.8	50.8	49.9

Table 8. 3–4 person matching accuracy (%). Independent = no history; Syn Test = with SignedVote.

	Independent	Syn Test
Greedy	30.4	46.3
Hungarian	30.8	31.1

Greedy versus Hungarian Table 8 compares Greedy and Hungarian assignment on the 3–4 person dataset. Without history, both strategies are near random (30.4% vs. 30.8%). With SignedVote accumulation, Greedy improves to 46.3% while Hungarian stagnates at 31.1%. This suggests that Greedy’s margin-based ordering is more compatible with historical noise than Hungarian’s global optimization.

Hard Threshold Table 9 evaluates the Hard Threshold strategy ($\tau = 0.7$, preserve action). Compared with the best per-fold SignedVote baseline (74.7%), Hard Threshold raises the mean to 79.5% (+4.8%). The gain is largest on Fold 3 (+18.6%) and Fold 4 (+14.9%), where intermittent motion noise is successfully filtered. Fold 2 degrades (–14.1%) because its margin-based confidence is unreliable, causing excessive freezing.

Learned Confidence Table 10 compares the learned confidence estimator against the two baselines. While it reaches 100.0% on Fold 4 and 99.4% on Fold 3, a catastrophic failure on Fold 1 (12.9%) drags the average to 66.3%, below both the SignedVote baseline (74.7%) and Hard Threshold (79.5%). The failure mode is attributed to the low-confidence action being implemented as *skip* (decay with-

Table 9. Hard Threshold ($\tau = 0.7$, preserve) vs. SignedVote baseline (%).

Fold	Baseline Best	Hard Threshold	Δ
1	100.0 (d=0.00)	100.0 (d=0.00)	0.0
2	49.7 (d=0.50)	35.6 (d=0.10)	-14.1
3	81.4 (d=0.00)	100.0 (d=0.00)	+18.6
4	67.6 (d=0.00)	82.5 (d=0.00)	+14.9
Mean	74.7	79.5	+4.8

Table 10. Learned confidence vs. baselines (%).

Fold	Baseline	Hard Thr.	Learned CE	Best
1	100.0	100.0	12.9	Baseline/HT
2	49.7	35.6	53.0	Learned
3	81.4	100.0	99.4	HT
4	67.6	82.5	100.0	Learned
Mean	74.7	79.5	66.3	HT

Table 11. Confidence-accuracy correlation (Spearman r), averaged over 4 folds.

Metric	Diag. Margin	Max Diag.	Softmax Margin	Entropy
Spearman r	0.854	0.414	0.849	-0.104

out adding new evidence) rather than *preserve* (fully freeze history). When confidence is systematically low, the score matrix decays to near zero.

Confidence-Accuracy Correlation Table 11 reports the Spearman correlation between four confidence metrics and per-frame correctness, averaged over the four folds. Diagonal Margin (0.854) and Softmax Margin (0.849) are the most stable signals, whereas Max Diagonal (0.414) and Entropy (-0.104) are less reliable.

6.4. Experimental Summary

Synthetic IMU. No single generation backend dominates both acceleration and gyroscope fidelity. `naive_kinematics` best preserves acceleration magnitude, while `wimusim` best preserves gyroscope dynamics. This motivates a hybrid synthetic stream that combines the acceleration column from the former with the gyroscope and quaternion columns from the latter.

Adaptive temporal sampling. The temporal-slicing results are dataset-dependent. On TotalCapture, `fixed_plus_energy_gl` achieves the best composite score (0.6978), suggesting that preserving fixed-grid coverage while adding high-motion windows improves matching. On the custom dataset, the fixed baseline

remains strongest by composite score (0.2533); adaptive oversampling appears to amplify noisy intervals in the smaller legacy subset.

History-based matching. SignedVote accumulation brings substantial gains on the 2-person merged data (70.0% at $\lambda = 0$ vs. 55.5% independent), but the optimal decay is session-dependent: pure accumulation helps Fold 1 (100.0%) yet hurts Fold 2 (30.8%), where erroneous early assignments must be forgotten. On unmerged tracks the gain is smaller (54.1% vs. 49.5%), confirming that identity fragmentation weakens temporal consistency.

In the 3–4 person setting, Greedy assignment with history rises to 46.3% while Hungarian stalls at 31.1%, because Greedy’s margin-based ordering can exploit the structured confidence landscape that SignedVote creates, whereas Hungarian’s global optimization locks in historical noise.

The Hard Threshold strategy ($\tau = 0.7$, preserve) raises the 2-person mean from 74.7% to 79.5% (+4.8%), with the largest gains on sessions afflicted by intermittent motion noise. The learned confidence estimator achieves perfect accuracy on two folds but suffers a catastrophic failure on Fold 1 (12.9%) because its low-confidence action was implemented as *skip* rather than *preserve*; its average (66.3%) therefore lags behind the simpler Hard Threshold. Finally, confidence-accuracy correlation analysis validates that Diagonal Margin (Spearman $r = 0.854$) and Softmax Margin ($r = 0.849$) are reliable signals for gating history updates, while Entropy is unstable across folds.

7. Discussions

7.1. Synthetic IMU Quality

Local frame alignment. During TotalCapture validation we found that synthetic acceleration axes periodically flip sign with respect to the real sensor. The right-arm `acc_x` from `naive_kinematics`, for example, is strongly anti-correlated ($\rho \approx -0.9$) before correction. The problem is not the world-frame acceleration—the second derivative of wrist position is physically correct—but the local coordinate frame used to rotate that acceleration into the sensor’s own axes. `naive_kinematics` rebuilds this frame on every frame with `build_frame`, inferring the x axis from the hand–wrist vector and the z axis from its cross product with the wrist–elbow vector. This treats the sensor as if its orientation were determined by instantaneous limb geometry, whereas a real IMU is fixed to the skin with a constant mounting rotation. When the arm swings, the inferred frame can jump 180°, flipping individual acceleration axes. `wimusim` avoids this because its frame is propagated from joint rotations rather than reconstructed from points. Existing synthesis methods therefore under-model the real

sensor mounting orientation. Future work should replace `build_frame` with a fixed sensor-to-bone rotation calibrated to the real placement.

7.2. Rendered Video Quality and Deployment Blockers

The rendered video pipeline described in Sec. 4.1.2 successfully produces synchronized RGB, mask, and bounding-box outputs, but several visual artifacts prevent the current renderings from being used as training data for the downstream Re-ID system. These artifacts fall into three categories that were observed during manual inspection of the generated sequences:

Ground penetration. Although a global height offset is applied to keep the minimum pelvis height above the ground plane, local penetration still occurs when a motion contains deep knee bends, crouching, or foot-planting that the pelvis-based offset does not capture. Actors can briefly clip through the floor, producing physically impossible silhouettes that confuse pose estimators trained on real-world data. A per-frame foot-contact solver or inverse-kinematics ground clamping would be needed to eliminate this class of artifact.

Floating and swimming motions. Some AMASS motions, especially those captured on treadmills or with sparse foot-contact labels, retarget to the SMPL-XL rig with an elevated or drifting root trajectory. The resulting rendered clips show actors gliding above the ground or performing locomotion without visible weight transfer, breaking the illusion of natural walking. Pre-filtering motions by foot-contact consistency, or adding a post-retargeting root-trajectory correction, would address this.

Mesh and clothing defects. The SynBody SMPL-XL assets include clothing geometry that can intersect with the body mesh or flip inside-out under extreme poses. Pants turning outward, jacket self-penetration, and shoulder-clipping were observed in sequences with large arm motions or torso twists. These defects create implausible texture patterns and silhouettes that are easy for human annotators to spot and likely degrade skeleton-extraction quality. Improving the clothing mesh topology, adding collision-aware garment simulation, or selecting motion-asset pairs with compatible pose ranges would reduce this issue.

Because of these artifacts, the rendered video clips produced in the current HuMoGen run were withheld from the training set. They were used only for qualitative inspection and pipeline debugging. Future work should treat each artifact class as a separate engineering target: ground-contact enforcement, motion quality filtering, and garment-mesh robustness. Once these are systematically addressed, the rendered video can be combined with the validated synthetic IMU streams to form a fully paired synthetic dataset for multi-person Re-ID training.

7.3. Why Adaptive Slicing Helps TotalCapture

The TotalCapture results suggest that adaptive slicing is most useful when it increases informative motion coverage without discarding the regular temporal grid. The best method, `fixed_plus_energy_g1`, keeps the fixed stride-16 baseline windows and adds a moderate number of high-motion windows. This likely improves training diversity while preserving stable coverage of the sequence. In contrast, pure motion-energy stride adaptation changes the sampling distribution more aggressively; it can improve Top-1, as in `motion_energy_g2`, but does not achieve the best composite score.

7.4. Why the Custom Dataset Behaves Differently

The custom dataset shows the opposite pattern: fixed windows remain best by composite score. Several factors may explain this. First, the custom legacy subset is small, so adaptive oversampling can amplify noisy or atypical intervals. Second, extracted skeleton quality and identity consistency may vary more across custom videos than in TotalCapture. Third, the current custom split reuses the existing project configuration, including its session-level split behavior, so the evaluation may be sensitive to a small number of sessions. Finally, the adaptive methods optimize temporal sampling density, but they do not directly solve detection noise, tracking identity switches, or IMU-person label ambiguity.

7.5. Metric Tradeoff

The custom results also show a useful tradeoff. `motion_energy_g1` obtains the strongest adaptive Top-1 score, but its grouped mean accuracy is substantially lower than fixed. This means that better single-window discrimination does not necessarily translate into robust multi-person matching when the candidate pool grows. For this project, grouped accuracy and composite score are therefore better indicators of practical pipeline performance than Top-1 alone.

7.6. Why Greedy Outperforms Hungarian with History

At first glance, Hungarian assignment should dominate Greedy because it is globally optimal for the per-frame matching objective (Eq. 9). Yet the experiments show the opposite: once history is accumulated, Greedy rises from 30.4% to 46.3% while Hungarian stagnates at 31.1% on the 3–4 person dataset (Table 8). This reversal has two complementary explanations.

History amplifies correct matches selectively. Signed-Vote accumulation (Eq. 5) reinforces pairs that are consistently similar across frames. For a correct IMU-track pair, the scores add constructively; for an incorrect pair, noise

tends to cancel out over time because the similarity fluctuates. Consequently, the accumulated matrix $\mathbf{H}^{(t)}$ develops a *structured* confidence landscape: a few entries become very large (correct matches), while the rest remain moderate or decay. This structure is exactly what Greedy exploits.

Greedy prioritizes high-confidence decisions. Greedy sorts candidates by the margin m_i (Eq. 8) and commits the most confident matches first. When history has amplified the correct entries, these high-margin decisions are overwhelmingly likely to be right, effectively removing the corresponding IMU and track from the pool before any ambiguous cases are considered. Hungarian, by contrast, optimizes the *sum* of all assigned scores. A single historically inflated but incorrect entry can force Hungarian to accept a sub-optimal global configuration in order to maximize that total. In other words, Hungarian treats every entry as equally trustworthy, whereas Greedy treats the margin as a reliability filter.

8. Conclusion

This report presents an end-to-end multi-person skeleton-IMU alignment pipeline and evaluates strategies for its two core obstacles: data scarcity and accumulated error.

On the data side, we find that no single synthetic IMU backend dominates both acceleration and gyroscope fidelity, motivating a hybrid generation route. Explicit signal-level validation remains essential before downstream use.

On the accumulated-error side, adaptive temporal sampling improves matching on large corpora but does not automatically generalize to smaller datasets. History-based decision making yields more consistent gains: SignedVote accumulation substantially raises per-frame accuracy, Greedy assignment outperforms Hungarian when historical noise is present, and a simple hard threshold strategy surpasses both the raw baseline and a learned confidence estimator that suffers from a fragile low-confidence action.

Overall, the results show that matching accuracy is determined not by the matcher alone but by the quality of upstream data, the informativeness of temporal windows, and the robustness of evidence accumulation.

9. Future Work

Future work should address four directions.

Learned confidence. The current estimator uses a *skip* action that decays history without adding new evidence, causing catastrophic failure when confidence is systematically low. Replacing skip with *preserve* (fully freeze history) or introducing a dynamic threshold that falls back to the hard threshold when learned confidence is uncertain could combine the stability of hand-crafted rules with the adaptivity of learned gating.

Real-time deployment. The current pipeline operates as an offline batch process: video and IMU data are first collected in full, then fed through detection, tracking, pose estimation, and matching in a centralized backend. For practical monitoring applications, the system must be restructured as an online stream-processing pipeline with bounded latency, where each incoming frame triggers incremental matching decisions rather than waiting for the entire sequence.

Synthetic data expansion. Rendered-video synthesis should be integrated with the hybrid IMU backend to produce fully paired training data, and IMU validation should be expanded to more body locations and larger corpora beyond the two arm-mounted sensors studied here.

End-to-end validation. The current report validates components in isolation, but a complete assessment should evaluate the tracker, pose estimator, matcher, and temporal aggregation both as separate modules and in the full end-to-end system. The broad synthetic IMU comparison is still a signal-level evaluation on two arm-mounted sensors; future work should cover more body locations, larger corpora, and downstream Re-ID performance when synthetic streams are mixed with rendered video. The current temporal-slicing implementation uses fixed input length and only changes sampling density; future work could tune adaptive budgets per dataset, add visibility-aware constraints, or learn sampling policies from validation performance. More robust conclusions would also require multiple seeds, larger custom evaluation sets, and multi-person data with more diverse camera viewpoints.

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